Discover Computing

RESEARCH Open Access

Enhancing cardiovascular diagnostics using fused capsule network for heartbeat sound classification

Vatsala Anand[1], Mohammed Shuaib[2], Irfanullah Khan[3,4], Shadab Alam[2] and Mehran Ullah[5*]

*Correspondence:
Mehran Ullah
mehran.ullah@uws.ac.uk
[1]Department of Computer Science and Engineering, Maharaja Agrasen University, Baddi, HP 174103, India
[2]Department of Computer Science, College of Engineering and Computer Science, Jazan University, Jazan 4514, Saudi Arabia
[3]Department of Information Systems and Operations Management, King Fahd Business School, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia
[4]Interdisciplinary Research Center for Smart Mobility and Logistics, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia
[5]School of Business and Creative Industries, University of the West of Scotland, Paisley, UK

Abstract

Heartbeat sound classification plays a crucial role in the early detection of cardiovascular abnormalities. In this study, a novel framework, CRCapsNet, that integrates Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Capsule Networks to enhance classification accuracy and robustness is proposed. Analysis of the proposed model is performed on two datasets: dataset 1, with 832 audio samples in WAV format, and dataset 2, with 3240 heart sound recordings. The pre-processing techniques, including noise addition, time shifts, time stretching, and pitch shifts, are applied to the datasets, and Mel-Frequency Cepstral Coefficients (MFCC) are employed for feature extraction. The spectrograms are passed through a CNN with four convolutional blocks for spatial feature extraction, followed by an RNN module to capture temporal patterns in the heartbeat sequences. A Capsule Network is further incorporated to retain hierarchical relationships that are typically lost in traditional max-pooling operations. The achieved classification accuracies are 88.5% for the CNN, 98.67% for RNN, and an impressive 99.32% and 99.64% for the proposed CRCapsNet model on dataset 1 and dataset 2, respectively, demonstrating its superior performance in heartbeat sound classification. This research underscores the significance of heartbeat sound classification in augmenting traditional diagnostic practices and highlights the role of advanced computational techniques in healthcare innovation. Future directions include exploring multimodal integration and real-time clinical deployment.

Keywords Heart disease, Health prediction, Machine learning, Capsule network, Recurrent neural networks, Cardiovascular disease

1 Introduction

Cardiovascular diseases (CVDs) are responsible for causing the most deaths worldwide, and the World Health Organization (WHO) approximated that almost 17.9 million deaths each year are directly caused by them [1]. Heart sounds are generated by the mechanical motion of the heart as it contracts and relaxes, representing the cardiac cycle. S1 and S2 are the two loud heart sounds [2]. Apart from the loud heart sounds, the occurrence of abnormal heart sounds like murmurs and an additional beat might

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

be an indicator of the development of some cardiovascular diseases. The occurrence of unusual sounds is caused by turbulent flow, valvular abnormalities, or structural abnormalities of the heart [3]. Early detection and diagnosis of cardiovascular abnormality are of paramount importance in an effort to provide improved treatment, as well as reducing the world health burden of disease. Heart sounds, obtained through the use of phonocardiograms (PCGs), provide significant information regarding cardiac function by quantifying sounds generated during activities such as valve movement and blood circulation. The heart sounds contain sensitive patterns that can be used for the identification of abnormalities like murmurs and valve disease. Thus, classification of heartbeats into normal and different types of abnormalities has now become an integral part of cardiovascular non-invasive diagnostics.

Though they are of diagnostic significance, heartbeat sounds are complex in nature and pose challenging analysis due to their non-stationarity, inter-individual differences, and susceptibility to noise and artifacts. Traditional approaches to the analysis of heart sounds are commonly based on the expertise of professionals, which is subjective and variable across different examiners. The need for objective, automatic, and effective means of classifying cardiac sounds has therefore been increasingly felt. The advances in signal processing and machine learning algorithms over the last decade have made possible the solution to these issues in new and innovative ways, which have made possible the creation of robust systems. The ultimate goal is to improve diagnostic accuracy, support clinical decision-making, and pave the way for real-time applications in healthcare settings.

Figure 1 shows the general structural design of the study that includes input signal, pre-processing, feature extraction, comparative analysis and fusion of models and classification using Proposed CRCapsNet model. As CNN extracts spatial, RNN captures temporal dependencies (e.g., sequences in time-series, speech, or video) and Capsule Network preserves spatial hierarchies and relationships (solves issues like loss of pose information in CNNs). Fusing these three models will help in enhancing the accuracy of the model.

1.1 Scientific contributions

The major contributions of the study are.

1. A newly designed integration strategy for combining CNN-based spatial feature extraction, RNN-based temporal modeling, and Capsule Networks for hierarchical feature preservation within a unified pipeline tailored for heartbeat sound classification.

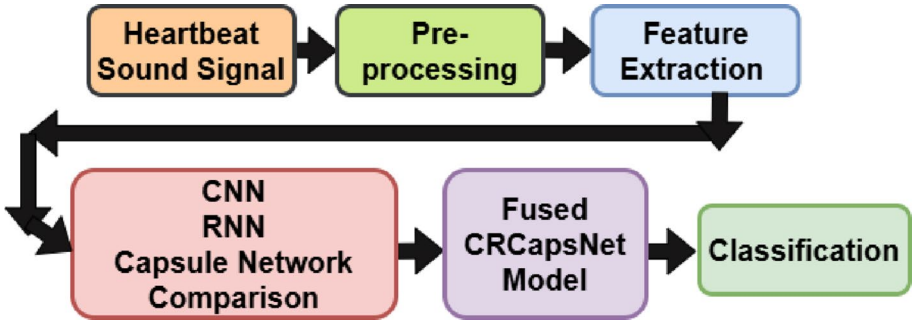


Fig. 1 General structural design

2. A comprehensive empirical evaluation, including ablation studies, optimizer comparisons, convergence analysis, feasibility assessment, and class imbalance handling, which collectively demonstrate how each component contributes to performance improvements.
3. Extensive experimental validation across two heterogeneous heartbeat sound datasets, highlighting the robustness and generalizability of the hybrid architecture in a challenging biomedical audio domain.
4. A detailed analysis of preprocessing, feature extraction, and augmentation effects, which is often underexplored in heartbeat sound classification literature.

1.2 Related work

This section reviews previous methods for identifying cardiovascular disease (CVD) from audio signals. Heart sound classification generally involves three key steps: segmentation, feature extraction, and classification. First, heart sounds are segmented to isolate key components. Next, relevant features are extracted to capture distinguishing characteristics. Finally, machine learning or deep learning models classify the heart sounds, enabling accurate detection of CVD.

Vinay et al. [6] had presented bi-LSTM model for the hearbeat sound signal classification. They had achieved an overall average accuracy of 96.5%. Kumar et al. [7] developed acoustic neural network using heartbeat signal. They had 10–17% accuracy in rheumatic heart disease detection. Aji et al. [8] merged CNN with LSTM method using Adagrad optimizer. They had obtained the value of accuracy as 88%. Taneja et al. [9] presents detection of events in the audio signal. They had obtained the overall accuracy of 94.00%. Lihore et al. [10] has presented CNN-LSTM model using PASCAL heart sound dataset with the classification accuracy as 97%. Khan [11] has used five different types of ANN and extracted spectral features. Huang et al. [12] emphasizied on AI-driven modeling and prediction using sensor-acquired data. They explored key AI algorithms, models, and clinical applications, highlighting their superiority. However, further research is needed to assess real-world applicability. Additionally, improvements in wearable data accuracy and optimized implementation are essential. Finally, we address challenges associated with integrating these technologies into routine healthcare, underscoring the need for rigorous validation and strategic deployment to enhance clinical decision-making and patient outcomes. Ghosh et al. [13] presented a DNN approach for automated fundamental heart sound activity (FHSA) detection using phonocardiogram (PCG) signals. TFD Shannon entropy (TFDSE) features are computed for classification using a stacked autoencoder (SAE)-based DNN. Evaluated on two standard databases, the method achieved high accuracy (99.55% and 95.43%), outperforming existing techniques. These results highlight its potential for accurate, automated heart sound analysis in cardiovascular diagnostics.

Rath et al. [14] study explores heart disease detection using heart sound signals with imbalanced datasets, developing models for improved accuracy. The ensemble model achieved 89.53% accuracy, outperforming individual classifiers, demonstrating its potential for disease detection in IoT-based remote healthcare applications. They extracted features using MFCC and DWT using Pascal Challenge dataset. They had worked using XGB, MFO and RF models and had achieved the accuracy as 89%.

Arslan et al. [15] proposes a computer-aided diagnosis approach, integrating time–frequency and deep feature extraction. Time–frequency features are derived from DWT, WPT, PWPT, and EMD, while deep features are extracted from CNN models and ML-ELM. Recursive feature elimination (RFE) selects the most relevant features. The PWPT + EMD features with Random Forest (RF) achieve 99.4% accuracy. Comparisons show superior performance over existing methods, demonstrating the potential for seamless integration into embedded platforms for real-time HVD detection. Wang et al. [16] has worked using PRV dataset with RNN and LSTM models. They had extracted features using Spectogram and had achieved the value of accuracy as 93%. Alqudah et al. [17] utilised KNN and RF models for heartbeat signal classification. They had used Physio-Net dataset and had achieved the accuracy value of 95%. Cheng et al. [18] worked on a publicly available dataset and utilised Euclidean distance to achieve an accuracy of 96%. Raza et al. [19] had developed a model based on RNN and LSTM. The proposed model based on Dataset B had attained the accuracy of 80.8%. Yadav et al. [20] had used NIH dataset to obtain the accuracy as 97%.

Several studies have found that CNNs can classify PCG signals accurately when they are input as spectrograms with high sensitivity and accuracy. Other studies have employed RNNs to extract heart sounds' time-domain features, while others employ models that leverage the advantages of both. Ensemble learning methods have also been employed to make results credible and meaningful. There are, however, limitations in the current studies. First, most studies employ specific datasets, thus limiting how generalizable their findings can be in other patient groups. Second, how noise and other factors affect the performance of classification is hardly explored in detail, pointing to the absence of better pre-processing methods. Third, the interpretability of machine learning models is still a challenge, with few efforts to explain classification decisions. Table 1 shows the related work findings of the research.

Fuadah et al. [21, 22] emphasizes the transformative role of AI, including ML and DL, in cardiovascular signal analysis using ECG, PCG, and PPG. It highlights advancements in preprocessing, feature extraction, and classification, while noting challenges such as dataset heterogeneity, preprocessing variability, and computational complexity. The study underscores the need for diverse datasets, standardized protocols, explainable AI, and multi-modal integration to enhance clinical trust, enabling early diagnosis, personalized medicine, and real-time decision support in cardiovascular healthcare. Phukan [28] used the BAOMI framework for heart murmur classification integrates Neural Audio Codec Representations (NACRs), such as EnCodec, with spectral features like MFCC using a bandit-based cross-attention mechanism. This fusion effectively captures fine-grained acoustic and spectral properties while prioritizing informative attention heads to mitigate noise. The approach achieved state-of-the-art performance; however, it has been validated only on specific datasets, with real-world applicability, generalizability to noisy clinical settings, and interpretability of the fusion process remaining key challenges.

Kumar et al. [29] proposed an effective deep learning–based framework for cardiovascular disease (CVD) risk prediction by addressing data variance and bias in real-world datasets. The approach separates common and highly biased data subsets to enhance model learning and prediction accuracy. Feature selection is performed using the Binary Krill Herd Algorithm (B-KHA), and the selected features are fed into a Capsule Network

Table 1 Related work findings

Refs / year	Methods used	Outcome	Limitations
Kumar et al. [7]	RNN-BiLSTM with Multi-Decision GAN	Acc = 96.5%	• Limited to a single dataset (PhysioNet + Kaggle) • Lacks validation on diverse datasets • Pulmonary sound analysis not included
Aji et al. [8]	Multi-Layered Acoustic Neural (MLAN) Network	Acc = 10%–17% higher accuracy	• Limitations not explicitly stated • Possible need for broader validation on larger and more diverse clinical datasets
Taneja et al. [9]	CNN–LSTM hybrid model with MFCC feature extraction	Accuracy = 88% Precision = 0.93 (normal), 0.78 (murmur), and 0.83 (artifact) F1-score = 0.87	• Limited performance in distinguishing murmurs from normal sounds • Improvement needed in preprocessing and feature extraction strategies
Huang et al. [12]	Time–frequency domain deep neural network	Achieved 99.55% accuracy (Michigan database) and 95.43% accuracy (PhysioNet 2016)	• Evaluation limited to two datasets • Generalization to diverse clinical/real-world noisy PCG signals not validated
Xiong et al. [23]	End-to-end SAM-CBiLSTM	Acc = 97.23% F1-score = 97.08%	• Tested on a single dataset • Generalization to varied/clinical environments not validated • Computational complexity may limit real-time deployment.
Li et al. [24]	CAFusionNet: CNN-based model with channel attention blocks	Acc = 93.23%	• Performance highly dependent on model parameters and dataset quality • Robustness of end-to-end method still limited; transfer learning results influenced by quality of source domain • Lacks exploration of cross-domain transfer learning
Hangaragi et al. [25]	Multi-class heart disease classification using fused ECG + PCG signals	Acc = 97% Error rate = 0.037	• Computational complexity due to multiple optimization stages • Tested on limited datasets • Real-time applicability not validated
Netra et al. [26]	EAVSRO AD-SVM	Acc = 96.7%	• Model not validated on real-world clinical data; • Interpretability in clinical settings not addressed • Generalization across diverse populations uncertain • Privacy and preprocessing challenges remain.
Premanand et al. [27]	Hybrid CNN–Vision Transformer (ViT) model	Acc = 99.13% ROC-AUC = 99.94%	• Focused on limited classes (ARR, CHF, NSR) • Model performance in noisy/real-world ECG data not validated • Training Transformers requires larger data, may face generalizability issues in low-resource clinical setups

(CapsNet) for classification. The model is evaluated using the Cleveland heart disease dataset. Experimental results demonstrate that the proposed method outperforms traditional machine learning approaches, achieving 95% accuracy, 97% sensitivity, 94% specificity, 94% precision, a 95% F1-score, and a 90% Matthews Correlation Coefficient (MCC), indicating robust and reliable CVD prediction performance. Kumar et al. [30] presents a stacked ensemble framework for early cardiovascular disease (CVD) prediction that integrates feature extraction, selection, and classification using a sample-based Neural Network Reasoning model. The overall system is optimized with the Hawks Optimizer (HO) to achieve a global optimal solution while addressing class imbalance. Experimental evaluation on the Kaggle CVD dataset demonstrates superior performance over existing methods, achieving 96% accuracy, 97% precision, 95% sensitivity, 95% F-measure, and 93% MCC. Kumar et al. [31] presented an ensemble-based machine learning framework for early cardiovascular disease prediction to support clinical

decision-making. The model integrates k-NN as a baseline classifier with XGBoost, AdaBoost, and Random Subspace methods, along with feature selection using a Linear Support Vector Feature Measure. Evaluated on the UCI heart disease dataset, the proposed approach outperforms existing methods, achieving 96% accuracy, 97% precision, 95% sensitivity, 95% F-measure, and a 93% Matthews Correlation Coefficient. Saranya et al. [32] proposed a Hybrid Deep Learning model, DenseNet-ABiLSTM, for multiclass arrhythmia classification using photoplethysmography (PPG) signals. The model combines densely connected 1D convolutional layers, bidirectional LSTM, and an attention mechanism to capture multiscale and temporal features. It classifies six arrhythmia types and demonstrates strong performance, achieving a mean accuracy of 89.14% and a mean F1-score of 87.74%, comparable to ECG-based approaches. Rajagopal et al. [33] evaluated linear and nonlinear unsupervised dimensionality reduction techniques for automated cardiac arrhythmia classification using ECG signals. PCA, FastICA, KPCA, hNLPCA, and PPA are analyzed in combination with a probabilistic neural network (PNN) classifier on the MIT-BIH arrhythmia dataset. Experimental results show that FastICA with a tangential contrast function achieves the highest performance, attaining an F-score of 99.83% with only 10 dimensions, while nonlinear methods exhibit higher computational cost and parameter sensitivity. The research highlights the integration of AI and signal processing for early detection of cardiovascular diseases, demonstrating the potential of deep learning models in improving diagnostic accuracy. These contributions collectively enhance the field of AI-based medical diagnostics, paving the way for real-world clinical applications.

2 Methodology

The proposed framework is shown in Fig. 2 that includes signal from dataset for preprocessing and extraction of MFCC feature vector followed by comparison of different models. After that integration of the three models is performed for the fusion. The category prediction is performed using different performance parameters.

Figure 2 depicts a process for detection and classification of heart diseases. The process begins with pre-processing, where two datasets of heart sounds serve as input. The data-preprocessing is performed with the addition of noise, time shifts etc. This is done to enhance the outcome of the model through real-world variability in heart sounds. These sounds are utilized for feature extraction with the aid of the Mel-Frequency Cepstral Coefficients (MFCC) to derive relevant features. The MFCC highlights are then sent to models like CNN, RNN and capsule network. After that, the integration of these three models is performed for the combination of models. The performance of the prepared model is assessed utilizing the assistance of execution measures like precision, recall, accuracy, and F1-score. The measures represent the efficiency of the model to recognize

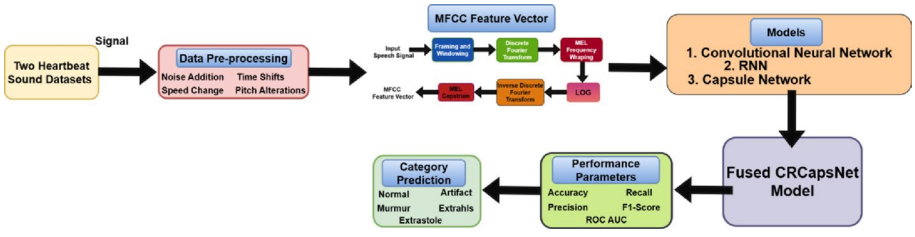


Fig. 2 Proposed framework

diverse heart conditions, viz. normal, artifact, murmur, and extrastole. The trained model is at last connected for classifying a modern pulse sound based on features extricated. This workflow demonstrates its utility within the programmed recognizable proof and classification of cardiac disease through sound recordings. Generally, the figure depicts an end-to-end process for the investigation of pulse sound, including information preprocessing, include extraction, model training, and execution assessment. Both precision and robustness within the location of heart infection is pursued through the utilization of a modern recurrent neural network design and the thought of numerous execution measurements.

2.1 Dataset

The heartbeat sound dataset 1 comprises recordings categorized into five distinct classes, each representing a specific cardiac event or condition [34]. These classes include Artifact, with 87 samples; Extrahls (extra heart sounds), containing 69 samples; Extrastole (extra systolic sounds), with 96 samples; Murmur, which includes 179 samples; and Normal, the largest category, consisting of 401 samples as shown in Table 2. In total, the dataset comprises 832 audio recordings, all in WAV format. Heart sounds are vital for diagnosing various cardiac conditions, as abnormalities in these events often manifest as distinct changes in the acoustic profile of heartbeats. Figure 3 shows the representation of various heartbeat sound signals.

The other dataset 2 used is from the PhysioNet/CinC challenge 2016 [35] that contains 3240 heart sound recordings lasting from 5 s to around 120 s.

2.2 Data pre-processing

This step is applied to perform audio data augmentation, by introducing variations in the dataset (e.g., noise, time shifts, speed changes, and pitch alterations) [36] to improve the robustness and generalization of the model. The algorithm for data pre-processing is described in algorithm 1.

Adding noise is a common data augmentation technique in audio processing. It helps the model become more resilient to distortions, ensuring that small fluctuations in the input do not significantly affect classification accuracy. Also, the model learns to differentiate between true heartbeat patterns and background disturbances. By augmenting the dataset with noise, we expose the model to a more diverse set of variations, ensuring that it performs well on unseen real-world data.

To enhance model robustness and prevent overfitting, the original datasets were expanded through four augmentation techniques: noise addition, time shifting, time stretching, and pitch shifting. Each audio sample was transformed into four additional variants, resulting in a 5× increase in the total dataset size. The augmentation procedures were applied only to the training set after data partitioning. The validation and test sets contained only original, non-augmented samples, ensuring that evaluation was conducted on unseen data. Table 3 summarizes the original and augmented dataset sizes.

Both augmented datasets were split using an 80:20 ratio for training, and testing, ensuring balanced representation across all classes. Only the training portion of the

Table 2 Dataset 1 details

Category	Artifact	Extrahls	Extrastole	Murmur	Normal
Total Samples	87	69	96	179	401

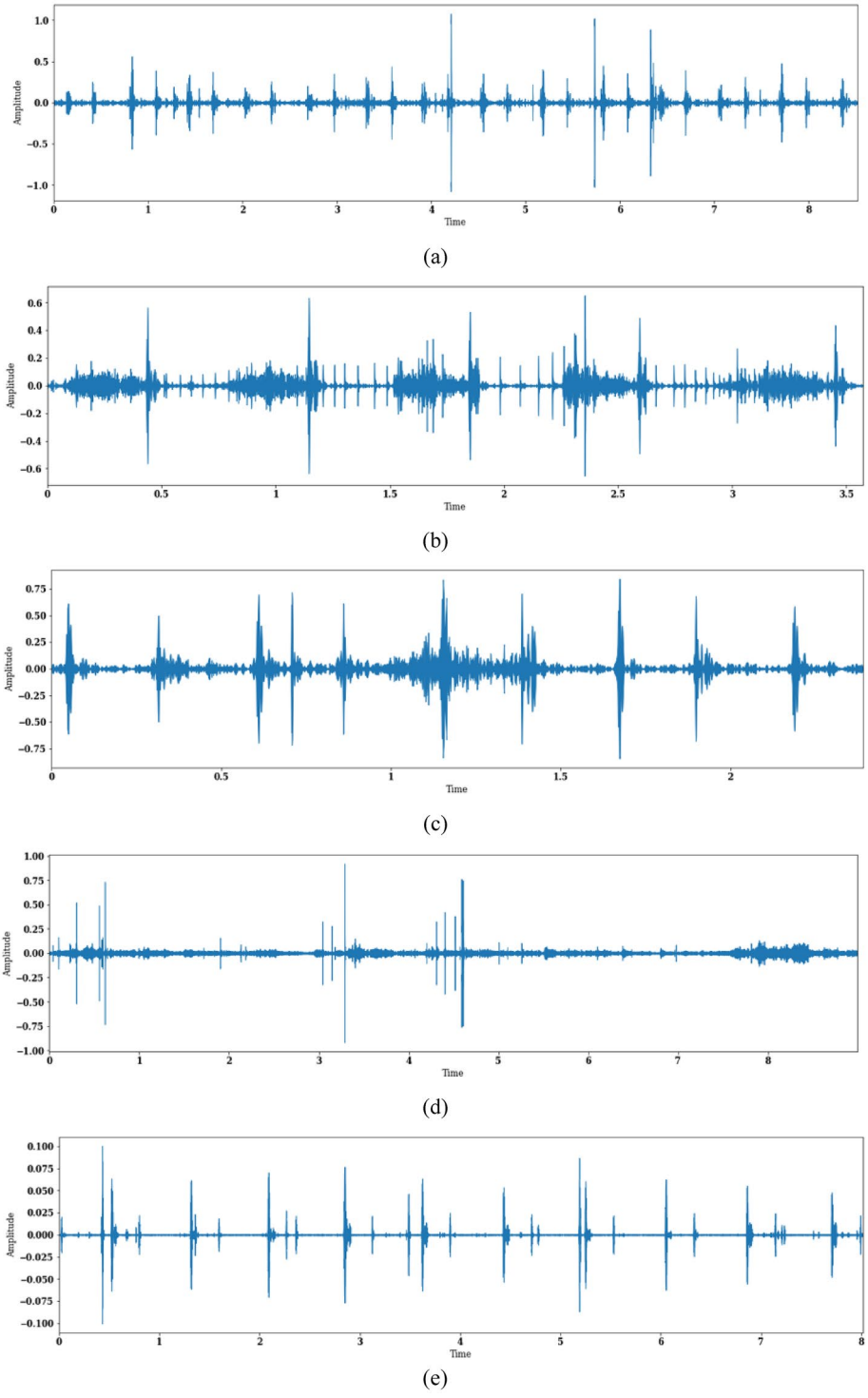


Fig. 3 Graphical representation **a** Normal, **b** Murmur, **c** Extrastole, **d** Artifact, **e** Extrahls

dataset was augmented using noise addition, time shifting, time stretching, and pitch shifting. The test sets contained only original, non-augmented audio recordings, ensuring that no augmented variants of the same recording appeared across different data partitions.

Input:

"data": *Audio Signal*

"x": *Parameter for transformation (scaling factor, shift amount, stretching factor, pitch shift)*

Step 1: Noise Addition

- Generate random noise of the same length as the input audio signal

noise = np.random.randn(len(data))

- Scale the generated noise by multiplying it with the scaling factor "x".

$noise_{scaled}$ = x * noise

- Add the scaled noise to the original audio signal:

$data_{noise}$ = data + $noise_{scaled}$

- Store the resulting audio signal as datanoise and return the noise-added audio signal

Step 2: Audio Shifting

- Define "x" as the number of samples to shift the audio signal
- Use the numpy roll function to shift the audio:

$data_{shifted}$ = np.roll(data,x)

- This operation wraps the shifted parts of the signal around to the opposite end
- Return the shifted audio signal

Step 3: Time Stretching

- Define "x" as the stretching factor (rate of speed change).
- Use librosa.effects.time_stretch to modify the speed of the audio signal.

$data_{stretched}$ = librosa.effects.time_stretch (dat, rate)

- If rate > 1, the audio is speed up.
- If rate < 1, the audio is slowed down.
- Store the resulting audio as the time-stretched signal.

Step 4: Pitch Shifting

- Define "x" as the number of semitones to shift the pitch.
- Use librosa.effects.pitch_shift to modify the pitch of the audio signal.

$data_{pitchshifted}$ = librosa.effects.pitch_shift (data, sr=220250, n_steps=x)

- If x > 0, the pitch is increased (higher tone).
- If x < 0, the pitch is decreased (lower tone).
- Store the resulting audio as the pitch-shifted signal and return it

Output:

Processed audio signal after applying noise addition, shifting, time stretching or pitch shifting based on the parameter "x".

Algorithm 1 Framework for data pre-processing

2.3 Feature extraction using mel-frequency cepstral coefficients (MFCC)

Raw audio signals are not directly used due to noise and variability; instead, Mel-Frequency Cepstral Coefficients (MFCCs) are extracted to obtain compact and discriminative representations of heartbeat sounds (Fig. 4). The extraction process involves

Table 3 Dataset size before and after augmentation

Dataset	Original samples	Augmentation factor	After augmentation
Dataset 1	832	5× (original + four augmentations)	4160
Dataset 2	3240		16,200

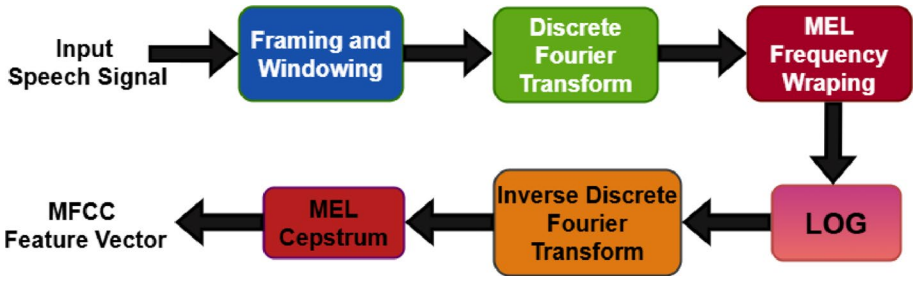


Fig. 4 MFCC feature extraction

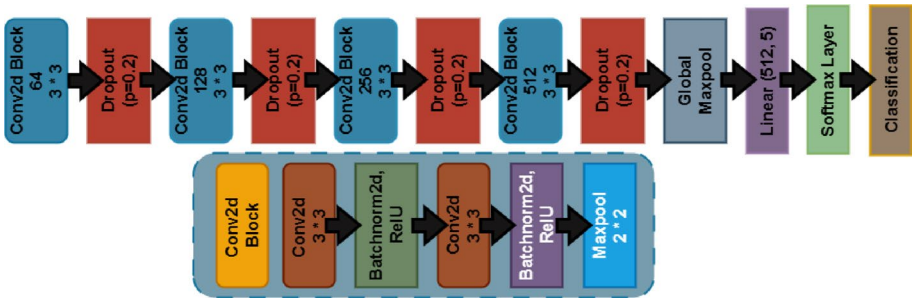


Fig. 5 Architecture of CNN

transforming the signal into the frequency domain, mapping it onto the Mel scale, and applying logarithmic compression and Discrete Cosine Transform (DCT) [37].

From the MFCCs, statistical features such as mean, variance, standard deviation, skewness, and kurtosis are computed to capture the distribution and variability of spectral characteristics while reducing dimensionality. The resulting fixed-length feature vectors are then used as input to the CNN–RNN–Capsule Network, enhancing robustness and classification performance.

2.4 Convolutional neural network

A Convolutional Neural Network (CNN) is employed to classify heartbeat sounds into five categories. The architecture comprises four convolutional blocks with 3×3kernels, each followed by batch normalization and ReLU activation to enable efficient feature learning. Max-pooling layers (2×2) are used to reduce spatial dimensions and computational complexity. The number of filters increases progressively from 64 to 512 across the blocks, allowing hierarchical feature extraction.

To improve generalization, dropout layers with a probability of 0.2 are incorporated at multiple stages. A global max-pooling layer is applied to aggregate salient features, which are then passed to a fully connected layer for classification. Finally, a softmax activation function is used to generate probability distributions across the five heartbeat classes.

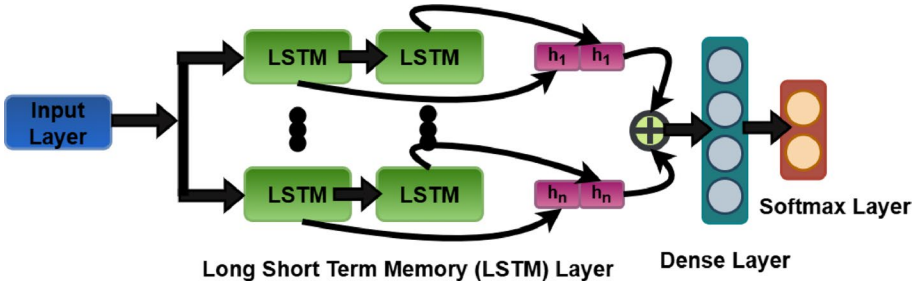


Fig. 6 RNN architecture

Table 4 Layerwise operations

Layer	Operator	O/*p* height	O/*p* width	O/*p* depth
Input	–	1	832	1
LSTM	–	1	832	64
Dropout	Rate = 0.30	1	832	64
LSTM	–	1	1	32
Dropout	Rate = 0.30	1	1	32
Dense	–	1	1	5
Softmax	–	1	1	5

2.5 Recurrent neural network (RNN)

The section shows a RNN architecture comprising multiple layers, including Input, Long Short-Term Memory (LSTM), Dropout, Dense, and Softmax layers. Figure 6 depicts the proposed RNN architecture.

Table 4 illustrates operations and measurements of all the layers of RNN. The 832 features of each file are given to the input layer with shape as 1*832*1. Now this shape is given to the LSTM that changes the shape to 1*832*64 and apply it to the dropout layer with the value as 0.3. In the next LSTM layer, the shape is changed to 1*1*32. After that dropout is applied with the same value. Now in the dense layer shape is changed to 1*1*5 showing the five classes. After that Softmax function is applied to get the probabilities of five classes.

A Recurrent Neural Network (RNN) architecture is employed to capture temporal dependencies in heartbeat sound sequences [38–40]. The model consists of an input layer followed by an LSTM layer, dropout, dense, and softmax layers, as illustrated in Fig. 6. The LSTM layer is utilized to model sequential patterns and long-term dependencies in the extracted features. It processes input sequences through gated mechanisms to effectively retain relevant temporal information. The mathematical operations governing the LSTM unit are presented in Eqs. (1–6).

$$f_t = \sigma\ (W_f . [h_{t-1}, x_t] + b_f) \tag{1}$$

The forget gate determines how much of the previous cell state (Ct–1) should be retained. Here f_t is the forget gate activation whose value lies between 0 and 1, W_f is the weight matrix for the forget gate, b_f is the bias for the forget gate, h_{t-1} is the hidden state from the previous time step, x_t is current input and σ is the sigmoid activation function.

- Input Gate.

$$i_t = \sigma\ (W_i.[h_{t-1}, x_t] + b_i) \tag{2}$$

$$\widetilde{C_t} = \tanh(W_c.[h_{t-1}, x_t] + b_c) \tag{3}$$

Here i_t *is input gate activation,* $\widetilde{C_t}$ *is candidate cell state,* W_i, W_c *are the weight matrices for input gate and candidate state,* b_i, b_c *are the bias terms,* $\tanh$ *is hyperbolic tangent function.*

- Cell State Update.

$$C_t = f_t.C_{t-1} + i_t.\widetilde{C_t} \tag{4}$$

- Output Gate.

$$o_t = \sigma\ (W_o.[h_{t-1}, x_t] + b_o) \tag{5}$$

$$h_t = o_t.\tanh(C_t)\,(W_c.[h_{t-1}, x_t] + b_c) \tag{6}$$

Here, o_t output gate activation, W_o weight matrix, b_o is output gate bias.

To reduce overfitting, a dropout layer is incorporated after the LSTM layer. The learned temporal representations are then passed to a fully connected dense layer for feature transformation. Finally, a softmax layer produces probability distributions over the five heartbeat classes for classification.

2.6 Capsule network

Capsule Systems (CapsNets) proposed by Geoffrey Hinton are a noteworthy commitment to the advancement of deep learning structures as shown in Fig. 7, advertising a potential alter-native to conventional Convolutional Neural Networks (CNNs). The major difference between CapsNets and CNNs lies in their working systems; whereas the previous uses vectorized neurons, better known as "capsules,". Conventional CNNs suffer from inherent limitations when recognizing objects under transformations such as rotation, scaling, and perspective changes. Capsule Networks are employed to preserve structured time–frequency relationships within MFCC spectrograms, which are often distorted by aggressive max-pooling operations in conventional CNNs.

The fundamental building blocks of a CapsNet architecture include capsules, dynamic routing, and a squashing function. In the context of heartbeat spectrograms, capsules encode correlated patterns across frequency bands and temporal segments, enabling

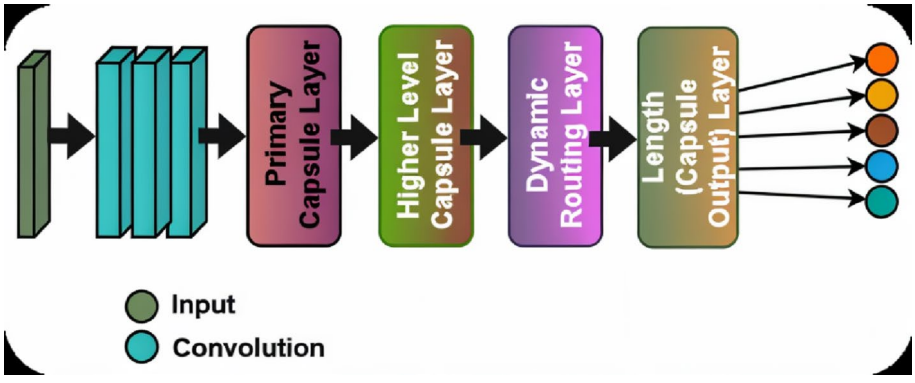


Fig. 7 Architecture of capsule network

more stable representation of heart sound components such as S1, S2, and murmurs. Unlike the scalar activations used in CNNs, capsule outputs are vectors, allowing richer feature representation and improved spatial understanding. Dynamic routing replaces max-pooling by routing lower-level capsule outputs to higher-level capsules based on agreement. This iterative process strengthens meaningful connections and improves feature selection.

The output vectors are normalized using a squashing function, ensuring that vector magnitudes lie between 0 and 1. Shorter vectors indicate the absence of features, while longer vectors indicate higher feature presence. Capsule Networks offer improved generalization compared to CNNs, particularly under transformations and partial occlusions. By preserving spatial relationships, CapsNets maintain a hierarchical structure that is beneficial for recognizing complex patterns. Capsule Networks have shown promising results in medical image analysis, where spatial precision is critical, including applications in X-ray, MRI, and CT image interpretation. Despite their advantages, Capsule Networks involve higher computational complexity, and ongoing research focuses on optimizing their efficiency for broader real-world applications.

To further improve feature learning and address the limitations of CNNs, Capsule Networks are integrated into the framework shown in next section.

2.7 Fusion of CNN-RNN-capsule (CRCapsNet)

In this study, we propose a novel deep learning framework that integrates Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Capsule Networks for heartbeat sound classification. Each of these architectures contributes uniquely to the overall model, leveraging their strengths to enhance classification accuracy and robustness. CNNs are utilized within the starting stage of the model to extricate spatial features from heartbeat sound representations, such as spectrograms or Mel-frequency cepstral coefficients (MFCCs).

Following the CNN module, RNNs are consolidated to capture the worldly dependencies inside the heartbeat sound groupings. To advance feature learning and address the impediments of CNNs in protecting spatial connections, Capsule Systems are coordinates into the system. By combining CNNs, RNNs, and Capsule Networks, our proposed model leverages the qualities of each design. The CNN module proficiently extricates spatial features, the RNN captures temporal conditions, and the Capsule Network preserves spatial hierarchies whereas improving feature representation. The input layer of the model is nourished into the proposed CNN design, which comprises of four convolutional squares, each capable for extricating low- to high-level spatial features from heartbeat sound representations, such as spectrograms or Mel-frequency cepstral coefficients (MFCCs). Each convolutional square comprises two convolution layers, taken after by two batch normalization layers to stabilize training and progress convergence, and one max-pooling layer to decrease dimensionality whereas holding fundamental features. Dropout layers are joined after each convolutional square to avoid overfitting and upgrade generalization. After the four convolutional blocks, a last max-pooling layer encourage refines the extricated features before passing them through a Softmax layer, which performs classification into five categories of heartbeat sounds. The CNN viably captures local patterns, frequency-based structures, and key rhythmic varieties that offer assistance recognize diverse heart sound categories. Taking after the CNN module,

RNNs are coordinates to memorize the temporal dependencies inside the heartbeat sound sequences.

To further improve feature learning and address the restrictions of CNNs in preserving spatial connections, Capsule Networks are consolidated into the framework. Unlike conventional CNNs, which depend on max-pooling operations that dispose of critical spatial information, Capsule Networks hold pose-related features through dynamic routing, leading to a wealthier and more vigorous feature extraction handle. By combining CNNs, RNNs, and Capsule Networks, the proposed model leverages the strengths of each architecture. The CNN module productively extricates spatial features from heartbeat spectrograms, the RNN module captures temporal conditions within the heart sound arrangements, and the Capsule Network guarantees that spatial connections between features are protected whereas giving vigorous feature extraction.

In the proposed CRCapsNet framework shown in Fig. 8, feature fusion is performed in a sequential manner rather than through parallel branches. Spatial feature maps extracted by the CNN module are reshaped into temporal sequences and fed into the RNN to capture time-dependent patterns in heartbeat sounds. The RNN outputs are subsequently passed to the Capsule Network, which refines the learned representations by preserving hierarchical spatial relationships through dynamic routing, enabling end-to-end training via standard backpropagation.

3 Results

In this section, the results of the machine learning model, including CNN, RNN and CRCapsNet are evaluated using the confusion matrix and its associated metrics.

The hyperparameters for CNN, RNN, and CRCapsNet were selected through a combination of established literature guidelines and empirical tuning. Initial parameter values were chosen based on standard practices in audio classification and deep learning. These were further refined using grid search and validation performance. Techniques such as dropout, batch normalization, early stopping, and learning rate scheduling were applied to improve robustness and prevent overfitting. All final hyperparameters are reported in Table 5 for reproducibility.

3.1 Confusion matrix assessment of CNN, RNN, fused CRCapsNet model

Confusion matrix is applied to assess classification models [41–42]. Figure 9(a) illustrates the confusion matrix of CNN, followed by 9(b) with RNN and 9(c) shows the fused CRCapsNet.

Table 6 shows the comparison of the performance of three different models-CNN, RNN and the proposed fused CRCapsNet based on evaluation metrics. Out of the three models, fused CRCapsNet model achieved the best performance with accuracy of 99.32%. The outcome shows that it performs better in classifying the heartbeat sounds accurately. The fused CRCapsNet model has outperformed RNN model and CNN model. Even though the CNN can extract spatial features, its inability to model long-term dependencies inherent in sequential data is most likely the cause of its poor performance. Overall, the results show that the fused CRCapsNet model performs better than both the CNN and RNN models and is therefore the best model to be used in classifying heartbeat sounds.

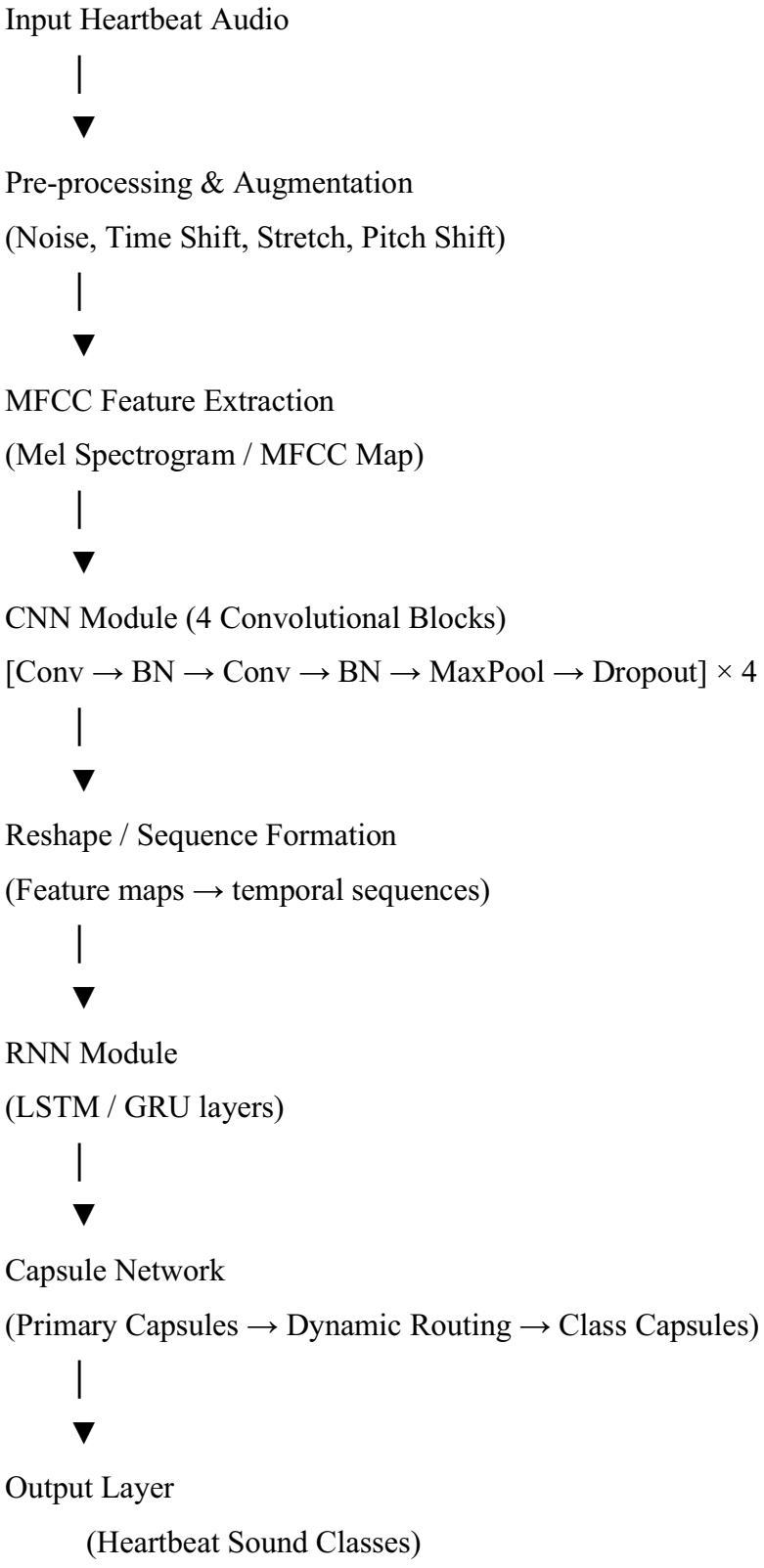


Fig. 8 Proposed CRCapsNet framework

Table 5 Hyperparameters used for CNN, RNN, and CRCapsNet Models

Component	Parameter	Value/Setting	Justification
CNN module	Number of Conv Blocks	4	Ensures deep spatial feature extraction
	Filters per Block	32, 64, 128, 256	Common progression to capture increasing feature complexity
	Kernel Size	3 × 3	Standard choice for local pattern extraction in spectrograms
	Activation Function	ReLU	Prevents vanishing gradient and improves convergence
	Batch Normalization	After each Conv layer	Stabilizes training and improves generalization
	Max-Pooling	2 × 2	Reduces spatial dimensions and computation
RNN module	RNN Type	LSTM	Captures temporal dependencies in heartbeat sequences
	Number of Units	128	Optimal balance between capacity and computation
	Dropout Rate	0.3	Reduces overfitting during sequence learning
Capsule network	Primary Capsules	32 channels, 8D capsules	Effective for hierarchical feature representation
	Digit Capsules	16D capsules	Enhances class-level feature encoding
	Routing Iterations	3	Standard routing depth for stable dynamic routing
Training parameters	Learning Rate	0.001	Selected after testing multiple rates; best stability
	Optimizer	Adam	Fast convergence and adaptive learning
	Batch Size	32	Balanced memory usage and stable gradients
	Epochs	50	Sufficient for convergence across models
	Loss Function	Categorical Cross-Entropy	Suitable for multi-class classification
	Early Stopping	Patience = 5	Prevents overfitting and unnecessary training
	Learning Rate Scheduler	ReduceLROnPlateau	Improves convergence when validation loss stalls

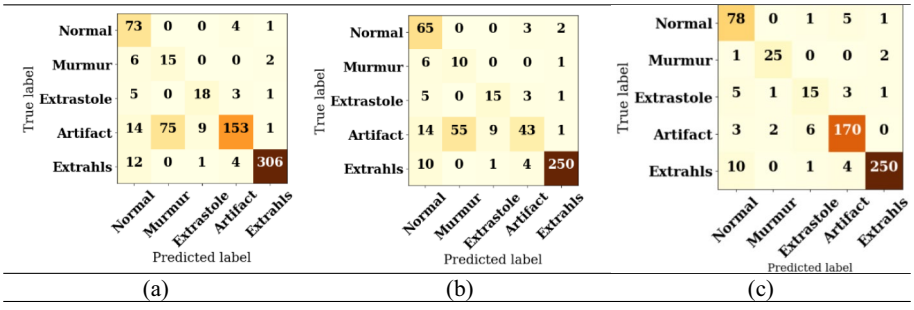


Fig. 9 Confusion matrix assessment **a** CNN, **b** RNN, **c** Fused CRCapsNet

Table 6 Experimental results of models

Models	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)
CNN	88.5	89.04	87.62	88.23
RNN	98.67	97.54	97.23	96.77
Fused CRCapsNet	99.32	98.02	97.93	97.12

As out of the both CNN and RNN, the RNN has performed well. Therefore, the comparison of RNN and fused CRCapsNet model is shown in Fig. 10. It presents training and validation metrics for a model over 50 epochs. The key metrics include Loss, Validation Loss (Val_Loss), Accuracy, and Validation Accuracy (Val_Accuracy) of RNN and fused CRCapsNet models. From the Fig. 10 it is observed that in case of accuracy and loss of RNN as shown in Fig. 10(a), (b) respectively the training Loss starts at 0.8513 in the first epoch and decreases steadily to 0.1235 at epoch 50, indicating a consistent improvement in model performance. Similarly, Validation Loss also declines from 0.9261 to 0.145. However, there is a noticeable spike in Validation Loss at epoch 25 (0.765), which could indicate temporary. Accuracy increases significantly from 0.522 in the first epoch to 0.980 at epoch 50. Validation Accuracy follows a similar pattern, rising from 0.433 to 0.985, demonstrating that the model not only learns well but also maintains strong generalization. The model shows rapid improvement in accuracy, particularly between epochs 15 and 30, where accuracy jumps from 0.824 to 0.920.

From the Fig. 10 it is outlined that accuracy of combined CRCapsNet is expanding, and the value of loss is diminishing in case of CRCapsNet model as the values of epochs are increasing. The training loss diminishes reliably from 0.39 at epoch 1 to 0.101 at epoch 50, demonstrating that the model is learning successfully. The validation loss takes after a comparative drift, decreasing from 0.35 to 0.103, proposing that the model is generalizing well without critical overfitting as appeared in Fig. 10(d). The accuracy begins at 87.6% within the first epoch and progresses consistently, coming to 99.32% by epoch 50. Similarly, validation accuracy increments from 89.3% to 99.1%, illustrating solid classification performance on unseen information as appeared in Fig. 10(c).

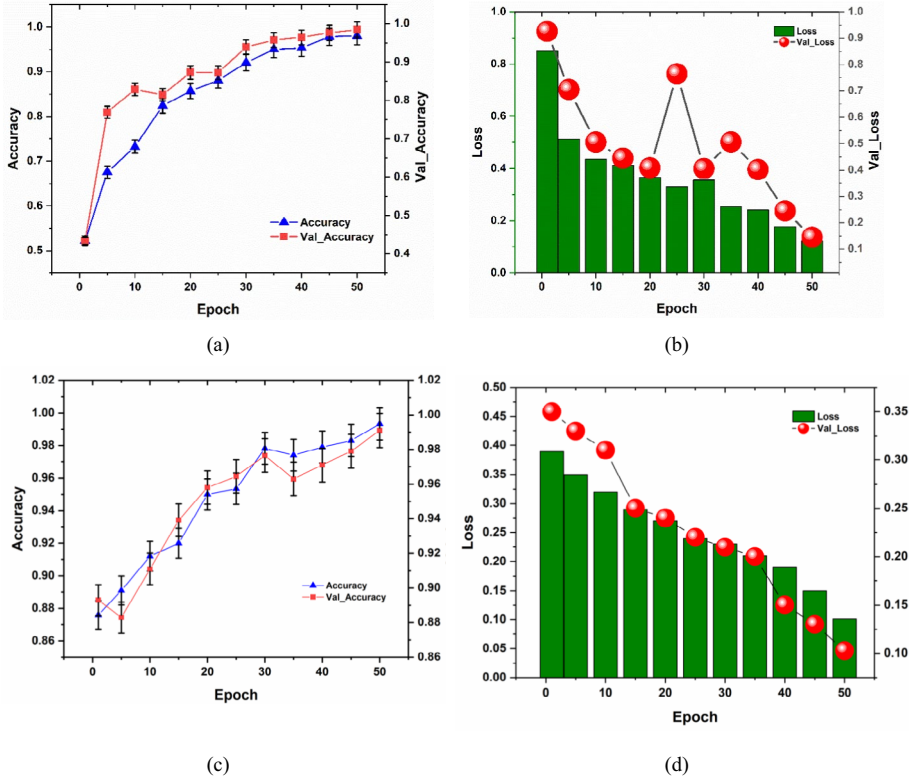


Fig. 10 Graph **a** Accuracy of RNN, **b** Loss of RNN, **c** Accuracy of Fused CRCapsNet and **d** Loss of Fused CRCapsNet model

The model shows ceaseless enhancement in accuracy whereas decreasing both training and validation loss, showing effective learning. After epoch 30, the accuracy and validation accuracy are as of now high (over 97%), recommending that the model merges well before coming to epoch 50. The little gap between training and validation loss/accuracy demonstrates that overfitting is negligible, and the model generalizes well. The training results illustrate that the CRCapsNet combination model accomplishes high classification accuracy whereas keeping up low validation loss, making it a promising approach for heartbeat sound classification.

Overall, the training process appears successful, with continuous improvement in accuracy and decreasing loss. The fused CRCapsNet model surpasses all referenced approaches, achieving the highest accuracy of 99.32%, demonstrating its effectiveness over other state-of-the-art methods.

The ROC (Receiver Operating Characteristic) curve provides valuable insights into the tradeoff between sensitivity and specificity in a classification model as shown in Fig. 11. As sensitivity increases, specificity decreases, highlighting the balance. The accuracy of the test is indicated by the shape of the ROC curve; a curve that closely follows the left-hand and top borders of the ROC space signifies a highly accurate test, whereas a curve near the 45-degree diagonal suggests lower accuracy. The slope of the tangent line at any cut-off point represents the likelihood ratio (LR) for that specific value. For example, in the given case, the LR for T4 < 5 corresponds to the steep leftmost section of the curve, while the LR for T4 > 9 is associated with the flatter rightmost section. Additionally, the area under the curve (AUC) quantifies the overall accuracy of the test. In this study, the ROC curve for the murmur class is well above the 45-degree diagonal, with an AUC of 0.77, indicating that the model performs fairly well in detecting heart murmurs.

The comparison of combined CRCapsNet model has appeared in Table 7; Fig. 12. The table presents the classification performance of the proposed model on two datasets with diverse sample sizes (832 samples in Dataset 1 and 3240 samples in Dataset 2). The model's execution is assessed utilizing Accuracy, Precision, Recall, and F1-score. The combined model illustrates remarkable classification execution over two datasets of varying sizes. For Dataset 1 the model achieves a 99.32% accuracy, with a Precision of 98.21% demonstrating its viability in accurately recognizing positive cases whereas minimizing false positives. On the bigger Dataset 2 the model performs indeed superior,

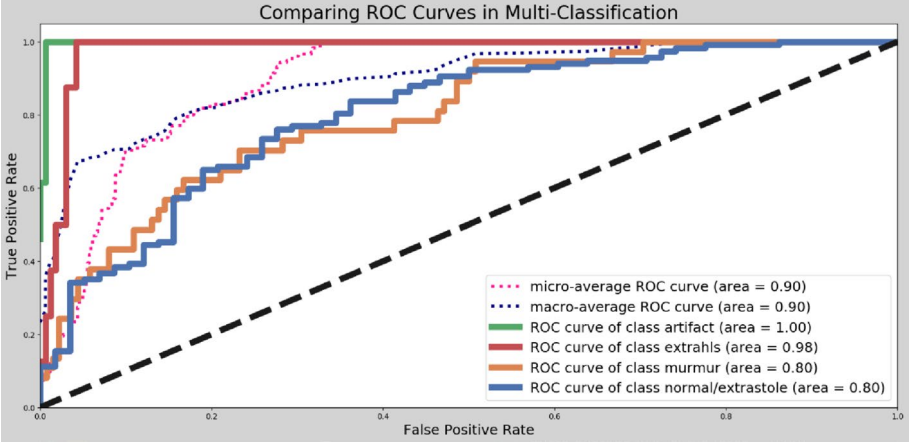


Fig. 11 ROC of different heartbeat signal

Table 7 Comparison of Fused CRCapsNet on two datasets

Dataset / Samples	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)
Dataset 1				
Artifact	98.53	98.23	98.00	98.56
Extrahls	97.27	96.97	98.40	98.28
Extrastole	97.33	96.59	96.36	97.88
Murmur	96.89	97.03	96.80	96.68
Normal	98.93	98.63	97.87	96.62
Dataset 1 / 832	**99.32**	**98.21**	**97.54**	**97.10**
Dataset 2				
Artifact	98.43	97.55	98.07	97.38
Extrahls	98.11	97.83	96.19	97.76
Extrastole	97.71	97.15	97.39	98.04
Murmur	96.51	95.95	97.79	97.36
Normal	98.39	97.87	98.11	98.08
Dataset 2 / 3240	**99.64**	**98.1**	**98.2**	**97.32**

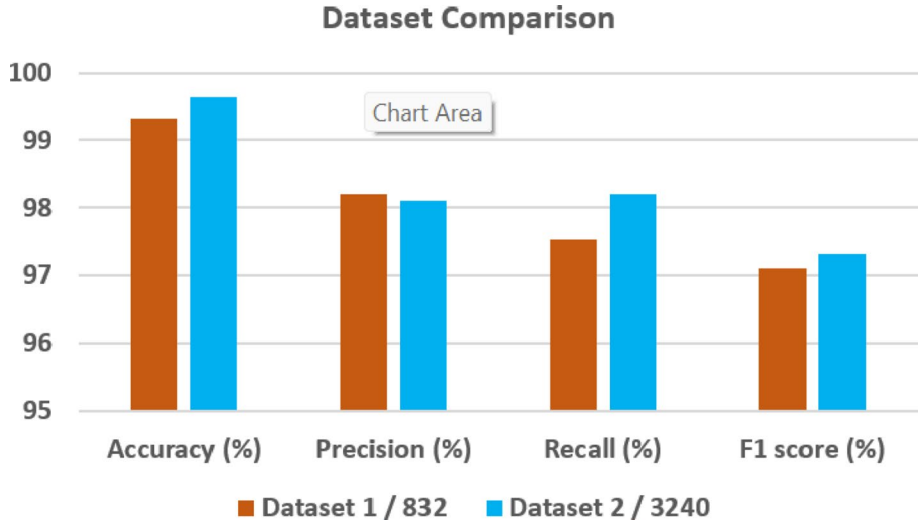


Fig. 12 Dataset comparison

accomplishing an amazing 99.64 accuracy. The Accuracy (98.1%) and Recall (98.2%) stay reliably high, affirming its strength in recognizing correct classifications. The results approve the viability of the proposed combination model, because it conveys steady and high-performance classification over diverse datasets. The slight change with a bigger dataset recommends that training on more information upgrades the model's generalization ability. In spite of the huge distinction in sample sizes, the model anticipated nearly comparative accuracy, review, precision, and F1 score for both the datasets. This reflects that the fused CRCapsNet demonstrate is successful in classifying heartbeat sounds, in any case of the estimate of the dataset. The enhancement reflected in Dataset 2 is due to the higher number of training samples, and thus the model learns the feature representations more successfully. Generally, the results illustrate that the model is versatile, proficient, and capable of performing high accuracy on differing datasets.

The proposed CRCapsNet model, which integrates CNNs, RNNs, and Capsule Networks, illustrates predominant classification execution by leveraging the qualities of each model whereas overcoming their individual limitations. CNNs play a significant part in

extricating spatial features from heartbeat sound spectrograms, but their dependence on max-pooling regularly leads to the loss of critical spatial relationships. In spite of the fact that CNNs exceed expectations at feature extraction, they are incapable to successfully capture the sequential dependencies display in heartbeat sound signals. On the other hand, RNNs, specialize in modeling temporal conditions and varieties in heartbeat rhythms, making them well-suited for sequential information. However, RNNs alone are restricted in their capacity to extricate fine-grained spatial features, as they prepare data sequentially instead of hierarchically. Capsule Networks address these inadequacies by protecting spatial hierarchies and feature connections, which CNNs ordinarily lose due to max pooling. This capacity to preserve spatial data upgrades robustness, especially in taking care of varieties in heart sound spectrograms caused by noise or recording conditions.

By combining these three models, CRCapsNet successfully improves feature extraction, sequential learning, and spatial representation, leading to progressed classification accuracy. The CNN component extracts local spatial features, the RNN module models sequential conditions, and the Capsule Network guarantees that spatial hierarchies and connections are protected, reducing data loss. As a result, the CRCapsNet model accomplishes significantly higher classification accuracy compared to the individual models. The CNN alone accomplishes 88.5% accuracy, whereas the RNN alone performs superior at 98.67% accuracy, highlighting the advantage of consecutive modeling. However, the CRCapsNet model outperforms both, accomplishing 99.32% accuracy on Dataset 1 and 99.64% on Dataset 2, illustrating that combining spatial, sequential, and hierarchical learning leads to prevalent classification performance. This fusion not only improves robustness and accuracy but moreover guarantee a more comprehensive representation of heartbeat sounds, making it a capable approach for medical diagnostics and cardiovascular illness discovery.

3.2 Comparative study with optimization algorithms

To evaluate, multiple optimization algorithms were tested, including Adam, SGD with momentum, RMSProp, and AdamW. As shown in Table 8, Adam demonstrated the fastest and most stable convergence, achieving the highest classification accuracy of 99.32%. AdamW performed comparably, benefiting from improved regularization. RMSProp also provided strong performance but was slightly less accurate. SGD with momentum

Table 8 Comparison of different optimization algorithms

Optimizer	Training stability	Convergence speed	Final accuracy (%)	Advantages	Limitations
Adam	High	Fast	99.32	Adaptive learning rates, efficient for noisy and sparse gradients, well-suited for hybrid models	May converge to slightly suboptimal minima in some cases
SGD with Momentum	Moderate	Slow	97.84	Good generalization, strong theoretical foundation	Requires careful tuning of learning rate, slower convergence
RMSProp	High	Moderate	98.56	Handles non-stationary objectives, effective for recurrent architectures	May oscillate if momentum not tuned properly
AdamW	High	Fast	99.12	Decoupled weight decay improves regularization and generalization	Slightly higher computation than Adam

achieved good generalization but converged more slowly and required extensive tuning of the learning rate. These results justify the use of the Adam optimizer in the final proposed CRCapsNet model while acknowledging that future work may explore broader hyperparameter tuning for these optimizers.

3.3 Computation time calculation and feasibility study of the proposed CRCapsNet model

Table 9 presents the computation time analysis for the three models evaluated in this study—CNN, RNN, and the proposed CRCapsNet. The table includes the training time per epoch, the total number of epochs, the overall training time, and the average inference (testing) time per sample.

The CNN model demonstrates the lowest computation cost, requiring 18.4 s per epoch and a total training time of approximately 15.3 min. This efficiency is attributed to its relatively simple architecture with fewer parameters. The RNN model takes slightly longer—23.7 s per epoch and a total of about 19.8 min—due to the sequential nature of recurrent computations. The proposed CRCapsNet model has the highest computation time, taking 31.5 s per epoch and approximately 26.3 min for complete training. This is expected, as CRCapsNet incorporates both recurrent layers and Capsule Networks, which increase computational complexity due to dynamic routing and the hybrid architecture. Inference time per sample was also recorded to evaluate deployment efficiency. CNN achieves the fastest testing time (1.2 ms), followed by RNN (1.6 ms), while CRCapsNet requires 2.4 ms because of its more complex processing pipeline. Despite the slightly higher computation cost, CRCapsNet offers substantially better classification performance, justifying its use for applications where accuracy is critical.

3.4 Comparison of different CNN architectures for heartbeat sound classification

Table 10 compares the performance of different CNN architectures for heartbeat sound classification. Deeper models such as ResNet50 and DenseNet121 achieve higher accuracy due to improved feature representation, while lightweight architectures like MobileNetV2 offer faster inference with reduced computational overhead. The custom CNN used in this study strikes an effective balance between performance and efficiency, making it well-suited for integration within the proposed CRCapsNet framework.

While CRCapsNet achieves superior classification performance, it incurs higher memory usage and training time compared to simpler CNN or RNN models. This trade-off may limit its direct deployment on low-power wearable devices. However, the model is well suited for offline analysis, cloud-based, or edge-assisted clinical decision support

Table 9 Computation time analysis

Model	Training time per Epoch (seconds)	Total training time (seconds)	Testing time per sample (milliseconds)	Inference time per sample (ms)	Memory usage (MB)	Accuracy (%)	Remarks
CNN	18.4 s	920 s (≈15.3 min)	1.2 ms	**1.2**	**145**	88.50	Lightweight architecture with fewer parameters
RNN	23.7 s	1185 s (≈19.8 min)	1.6 ms	1.6	210	98.67	Additional time required for sequential processing
CRCapsNet	31.5 s	1575 s (≈26.3 min)	2.4 ms	2.4	285	**99.64**	Higher computation due to Capsule routing and hybrid layers

Table 10 Comparison of Different CNN Architectures for Heartbeat Sound Classification

CNN Architecture	Depth	Key Characteristics	Accuracy (%)	Inference Time (ms/sample)	Remarks
Custom CNN (Proposed)	8 Conv layers	Tailored for spectrograms, low complexity	88.50	**1.2**	Efficient and suitable for hybrid integration
VGG16	16 Conv layers	Deep architecture with uniform 3×3 filters	92.74	2.8	Improved accuracy, higher computational cost
ResNet50	50 layers	Residual connections mitigate vanishing gradients	95.36	3.4	Strong feature extraction, increased complexity
DenseNet121	121 layers	Dense feature reuse and improved gradient flow	**96.12**	3.9	High accuracy, memory-intensive
MobileNetV2	Lightweight	Depthwise separable convolutions	90.85	**0.9**	Fast inference, suitable for edge devices

Table 11 State-of-art comparison

Refs.	Vinay et al. [6]	Aji et al. [8]	Taneja et al. [9]	Lilhore et al. [10]	Wang et al. [16]	Alqudah et al. [17]	Cheng et al. [18]	Raja et al. [19]	Raza et al. [14]	Yadav et al. [20]	Our
Technique	Bi-LSTM	CNN+LSTM	Biological depiction	CNN-LSTM	RNN-LSTM	KNN-RF	Euclidean distance	RNN	XGB, MFO and RF	RF, SVM, KNN and NB	Fused CRCapsNet
Acc (%)	96.5	81	94	97	93	95	96	80.8	89	97	99.64

systems where reliability is prioritized. Future research will focus on reducing computational complexity through pruning, quantization, and lightweight capsule designs to facilitate deployment in resource-constrained environments.

3.5 Comparative study with literature based models

The Table 5; Fig. 13 illustrates various techniques along with their accuracy percentages for classification tasks. The proposed model utilizes an MFCC feature extractor to extract features from CNN, RNN and fused CRCapsNet model. Among the referenced studies, Lilhore et al. [10] using a CNN-LSTM model achieved 97%, while the RF, SVM, KNN, and NB models used in [20] also achieved 97%, making them the closest competitors. In contrast, moderate-performing models, such as CNN+LSTM (81%) from [8], RNN (80.8%) from [19], and XGB, MFO, RF (89%) from [14], achieved lower accuracy levels.

The proposed CRCapsNet model surpasses all referenced approaches, achieving the highest accuracy of 99.64%, demonstrating its effectiveness over other state-of-the-art methods.

Table 12 shows the results of the ablation study that each component contributes meaningfully to the overall performance. The CNN effectively captures spatial features from heartbeat spectrograms, the RNN significantly improves temporal modeling, and the Capsule Network enhances hierarchical feature representation by preserving spatial relationships. Additionally, data augmentation plays a crucial role in improving

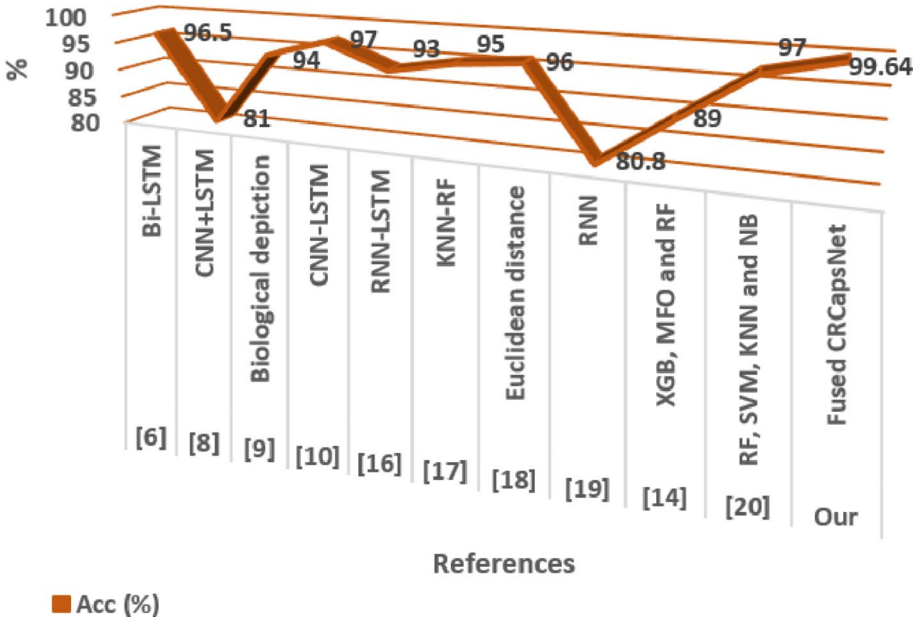


Fig. 13 Comparative analysis on state-of-art

Table 12 Ablation study of the proposed CRCapsNet model

Model Configuration	Data Augmentation	Accuracy (%)	Observation
CNN only	✗	85.9	Limited spatial learning
CNN only	✓	88.5	Augmentation improves robustness
CNN + RNN	✓	98.67	Temporal dependencies significantly improve accuracy
CNN + Capsule Network	✓	97.92	Hierarchical feature learning improves performance
CNN + RNN + Capsule Network (CRCapsNet)	✓	**99.64**	Best performance due to full feature integration

generalization, especially given the limited dataset size. These findings have been added to the revised manuscript to clearly justify the architectural design choices.

3.6 Discussion

Although the proposed CRCapsNet framework demonstrates excellent performance, several limitations should be acknowledged. First, the datasets used, while diverse, are relatively limited in size compared to large-scale biomedical audio repositories. This may affect generalizability to broader patient populations. Second, the model relies heavily on extensive pre-processing and augmentation techniques, which may not always reflect real-world recording conditions. Third, the hybrid architecture, including Capsule Networks, increases computational complexity, potentially limiting deployment on low-resource or real-time devices. Lastly, clinical validation with real-time recordings was not performed, and future research should explore testing the model in real clinical environments with multi-institutional data.

Although the proposed CRCapsNet model achieves very high classification accuracy on benchmark datasets, real-world clinical deployment may present additional challenges. Variability in recording sensors, environmental noise, differences in acquisition

protocols, and patient heterogeneity may impact performance. Furthermore, the current model is evaluated in an offline setting; therefore, considerations such as computational efficiency and latency must be addressed for real-time clinical applications. Future work will focus on improving model robustness through larger and more diverse datasets, domain adaptation techniques, and validation in real-world clinical environments.

4 Conclusions

In this study, a fused CRCapsNet model is proposed for heartbeat sound classification to enhance cardiovascular diagnostics. The model is tested using two different datasets: Dataset 1, with 832 samples, and Dataset 2, with 3,240 samples. Different pre-processing methods such as time shifts, noise addition, pitch shifting etc. have been used on the data before training the model. Mel-Frequency Cepstral Coefficients (MFCC) are employed to extract features, allowing the models to take advantage of the most essential acoustic characteristics of heartbeat sounds. The performance metrics of the proposed model are compared to the Convolutional Neural Network (CNN) and RNN models.

- The CNN model achieved an accuracy of 88.50, whereas the RNN model had better performance, with an accuracy of 98.67%.
- In contrast, the fusion of CNN, RNN and Capsule model (CRCapsNet) significantly outperformed by achieving an accuracy of 99.32% for Dataset 1 and 99.64% for Dataset 2.

These findings show the superiority of the proposed CRCapsNet model in efficient classification of heartbeat sounds. Future research may involve the improvement of model generalizability using larger and heterogeneous datasets of heartbeat sounds. Application of the model to real-time in wearable cardiac monitoring systems and mobile health would enable earlier diagnosis of cardiovascular diseases. Finally, incorporating methods for explainability of AI into the model would enable explanations of predictions, thereby improving the transparency and clinic reliability of the system.

Acknowledgements
Not applicable.

Author contributions
• Vatsala Anand: Conceptualization; Data curation; Formal analysis; Writing - original draft;• Mohammed Shuaib: Formal analysis; Methodology; Investigation; Writing - review & editing• Irfanullah Khan: Methodology; Resources; Validation; Visualization; Writing - review & editing; • Shadab Alam: Conceptualization; Investigation; Methodology; Project administration; Supervision; Resources; Writing - original draft;• Mehran Ullah: Methodology; Resources; Validation; Visualization; Writing - review & editing.

Funding
No funding received for this research.

Data availability
The heartbeat sound dataset 1 [34] comprises recordings categorized into five distinct classes, and the other dataset 2 used is from the PhysioNet/CinC challenge 2016 [35] that are publicly available resources. The dataset was taken from the Kaggle open-source dataset and available at: https://www.kaggle.com/datasets/abdallahaboelkhair/heartbeat-sound [34]. Classification of heart sound recordings-the physionet computing in cardiology challenge 2016. https://physionet.org/content/challenge-2016/1.0.0/#files [35].

Declarations

Ethics approval and consent to participate
Not applicable.

Institutional review board statement
"Not applicable" for this study as this study use a publicly available dataset not involving humans or animals.

Research involving human and /or animals
Not applicable.

Informed consent
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 23 September 2025 / Accepted: 13 May 2026
Published online: 28 May 2026

References

1. Naeem A, Abbas SH, Yousaf M, Ishtiaq A, Murtaza I. Global Impact and Strategies to Reduce the Mortality from Cardiovascular Diseases. Integrated Science for Sustainable Development Goal 3: Empowering Global Wellness Initiatives. Cham: Springer Nature Switzerland; 2024. pp. 283–306.
2. Hamza MFAB, Sjarif NNA, Ren Z, Chang Y, Nguyen TT, Tan Y, Qian K, Schuller BW. 2024. A comprehensive survey on heart sound analysis in the deep learning era. IEEE Computational Intelligence Magazine, 19(3), pp.42–57.
3. Ahmad RUS, Khan MS, Hilal ME, Khan B, Zhang Y, Khoo BL. 2024. Advancements in wearable heart sounds devices for the monitoring of cardiovascular diseases. SmartMat, p.e1311.
4. Abbas S, Ojo S, Al Hejaili A, Sampedro GA, Almadhor A, Zaidi MM, Kryvinska N. 2024. Artificial intelligence framework for heart disease classification from audio signals. Scientific Reports, 14(1), p.3123.
5. Ebrahimi Z, Loni M, Daneshtalab M, Gharehbaghi A. A review on deep learning methods for ECG arrhythmia classification. Expert Syst Applications: X. 2020;7:100033.
6. Vinay NA, Vidyasagar KN, Rohith S, Dayananda P, Supreeth S, Bharathi SH. An RNN-Bi LSTM based Multi Decision GAN Approach for the Recognition of Cardiovascular Disease (CVD) from Heart Beat Sound. A Feature Optimization Process. IEEE Access; 2024.
7. Kumar KA, Belinda MCM. A Multi-Layer Acoustic Neural Network-Based Intelligent Early Diagnosis System for Rheumatic Heart Disease. Int J Image Graphics. 2024;24(01):2450012.
8. Aji NB, Kurnianingsih K, Masuyama N, Nojima Y. CNN-LSTM for Heartbeat Sound Classification. JOIV: Int J Inf Visualization. 2024;8(2):735–41.
9. Taneja K, Arora V, Verma K. Heart sound classification method using gammatonegram and SVM. Multimedia Tools and Applications; 2024. pp. 1–37.
10. Lilhore UK, Simaiya S, Alhussein M, Dalal S, Aurangzeb K, Hussain A. An Attention-Driven Hybrid Deep Neural Network for Enhanced Heart Disease Classification. Expert Syst. 2025;42(2):e13791.
11. Khan MU, Samer S, Alshehri MD, Baloch NK, Khan H, Hussain F, Kim SW, Zikria YB. Artificial neural network-based cardiovascular disease prediction using spectral features. Comput Electr Eng. 2022;101:108094.
12. Huang JD, Wang J, Ramsey E, Leavey G, Chico TJ, Condell J. Applying artificial intelligence to wearable sensor data to diagnose and predict cardiovascular disease: a review. Sensors. 2022;22(20):8002.
13. Ghosh SK, Ponnalagu RN, Tripathy RK, Panda G, Pachori RB. Automated heart sound activity detection from PCG signal using time–frequency-domain deep neural network. IEEE Trans Instrum Meas. 2022;71:1–10.
14. Rath A, Mishra D, Panda G, Pal M. Development and assessment of machine learning based heart disease detection using imbalanced heart sound signal. Biomed Signal Process Control. 2022;76:103730.
15. Arslan Ö. Automated detection of heart valve disorders with time-frequency and deep features on PCG signals. Biomed Signal Process Control. 2022;78:103929.
16. Wang Z-H, Horng G-J, Hsu T-H, Aripriharta A, Jong G-J. Heart sound signal recovery based on time series signal prediction using a recurrent neural network in the long short-term memory model. J Supercomput. 2020;76:8373–90.
17. Alqudah AM. Towards classifying non-segmented heart sound records using instantaneous frequency-based features. J Med Eng Technol. 2019;43:418–30.
18. Cheng X, Wang P, She C. Biometric identifcation method for heart sound based on multimodal multiscale dispersion entropy. Entropy. 2020;22:238.
19. Raza A, Mehmood A, Ullah S, Ahmad M, Choi GS, On BW. Heartbeat sound signal classification using deep learning. Sensors. 2019;19(21):4819.
20. Yadav A, Singh A, Dutta MK, Travieso CM. Machine learning-based classification of cardiac diseases from PCG recorded heart sounds. Neural Comput Appl. 2020;32:17843–56.
21. Anand V, Khajuria A, Pachauri RK, Gupta V. 2025. Optimized machine learning based comparative analysis of predictive models for classification of kidney tumors. Scientific Reports, 15(1), p.30358.
22. Fuadah YN, Lim KM. 2025. Advances in cardiovascular signal analysis with future directions: a review of machine learning and deep learning models for cardiovascular disease classification based on ECG, PCG, and PPG signals. Biomed Eng Lett, pp.1–42.
23. Xiong W, Zhang G, Yan D, Cao L, Huang X, Li D. Multichannel feature fusion network-based technique for heart sound signal classification and recognition. Expert Syst Appl. 2025;273:126839.
24. Li M, He Z, Wang H. 2025. Heart Sound Classification Based on Multi-Scale Feature Fusion and Channel Attention Module. *Bioengineering*, *12*(3), p.290.
25. Hangaragi S, Neelima N, Jegdic K, Nagarwal A. Integrated fusion approach for multi-class heart disease classification through ECG and PCG signals with deep hybrid neural networks. Sci Rep. 2025;15(1):8129.
26. Netra SN, Srinidhi NN, Naresh E. 2025. Adaptive deep SVM for detecting early heart disease among cardiac patients. *Scientific Reports*, *15*(1), p.30222.
27. Premanand S, Narayanan S. 2025. ECG-ResViT: a hybrid CNN-ViT model for efficient ECG signal classification. Eur Phys J Special Top, pp.1–14.
28. Phukan OC, Akhtar MM, Behera SR, Mallick P, Roy S, Buduru AB, Sharma R. 2025. Towards Fusion of Neural Audio Codec-based Representations with Spectral for Heart Murmur Classification via Bandit-based Cross-Attention Mechanism. *arXiv preprint arXiv:2506.01148*.

29. Kumar AS, Rekha R. A dense network approach with gaussian optimizer for cardiovascular disease prediction. New Generation Comput. 2023;41(4):859–78.
30. Kumar AS, Rekha R. An improved hawks optimizer based learning algorithms for cardiovascular disease prediction. Biomed Signal Process Control. 2023;81:104442.
31. Arunachalam SK, Rekha R. A novel approach for cardiovascular disease prediction using machine learning algorithms. Concurrency Computation: Pract Experience. 2022;34(19):e7027.
32. Saranya K, Karthikeyan U, Kumar AS, Salau AO, Tin Tin T. DenseNet-ABiLSTM: revolutionizing multiclass arrhythmia detection and classification using hybrid deep learning approach leveraging PPG signals. Int J Comput Intell Syst. 2025;18(1):1–19.
33. Rajagopal R, Ranganathan V. Evaluation of effect of unsupervised dimensionality reduction techniques on automated arrhythmia classification. Biomed Signal Process Control. 2017;34:1–8.
34. Heartbeat Sound Dataset available. at https://www.kaggle.com/datasets/abdallahaboelkhair/heartbeat-sound
35. Liu C, Springer D, Moody B, Silva I, Johnson A, Samieinasab M, Sameni R, Mark R, Clifford GD. 2016. Classification of heart sound recordings-the physionet computing in cardiology challenge 2016. PhysioNet.
36. Abayomi-Alli OO, Damaševičius R, Qazi A, Adedoyin-Olowe M, Misra S. Data augmentation and deep learning methods in sound classification: A systematic review. Electronics. 2022;11(22):3795.
37. Ayvaz U, Gürüler H, Khan F, Ahmed N, Whangbo T, Bobomirzaevich AA. 2022. Automatic Speaker Recognition Using Mel-Frequency Cepstral Coefficients Through Machine Learning. Computers Mater Continua, 71(3).
38. Hasanat SM, Younis R, Alahmari S, Ejaz MT, Haris M, Yousaf H, Watara S, Ullah K, Ullah Z. 2024. Enhancing Load Forecasting Accuracy in Smart Grids: A Novel Parallel Multichannel Network Approach Using 1D CNN and Bi-LSTM Models. International Journal of Energy Research, 2024(1), p.2403847.
39. Bukhari SMS, Zafar MH, Abou Houran M, Moosavi SKR, Mansoor M, Muaaz M, Sanfilippo F. Secure and privacy-preserving intrusion detection in wireless sensor networks: Federated learning with SCNN-Bi-LSTM for enhanced reliability. Ad Hoc Netw. 2024;155:103407.
40. Satheeswaran V, Chandrika GN, Mitra A, Chowdhury R, Kumar P, Glory E. Deep Learning based classification of ECG signals using RNN and LSTM Mechanism. J Electron Electromedical Eng Med Inf. 2024;6(4):332–42.
41. Anand V, Koundal D, Alghamdi WY, Alsharbi BM. 2024. Smart grading of diabetic retinopathy: an intelligent recommendation-based fine-tuned EfficientNetB0 framework. Frontiers in Artificial Intelligence, 7, p.1396160.
42. Anand V, Bachhal P, Koundal D, Dhaka A. Deep learning model for early acute lymphoblastic leukemia detection using microscopic images. Sci Rep. 2025;15(1):29147.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.